Py

PyCon US
POSTER SESSION

RESULTS AT A GLANCE

Bigger ≠ Better
Scaling model size often raises energy, latency, and VRAM faster than toy-task quality.

Quantization Changes Economics
q4/q8 variants can cut memory and energy enough to change where mo

OVERVIEW

Everyday LLM calls look simple from Python, but the real deployment decision is a tradeoff between **quality**, **latency**, **VRAM**, **energy**, and **cost**.
We benchmark code generation, summarization, chat, RAG-style search, and embeddings across model sizes, quantization levels, architectures, and CPU/GPU configurations.

> **Thesis:** the most accurate LLM is often not the most economically efficient one. LLM deployment is a systems engineering problem, not just model selection.

BENCHMARK SCOPE

19 models / variants

5 workload families

234 metric rows

L40S NVIDIA L40S

Question	Measured Signals	Why It Matters
Bigger model?	toy score, latency, joules, VRAM	find diminishing returns
Quantize?	q4/q8/fp16, tokens/J	fit smaller hardware
Which workload?	task score by model	avoid one-model thinking

MEASUREMENT PIPELINE

Each run produces one row of metrics: latency slices, token counts, active-device energy, memory, and workload score.

FIXTURES	Repeatable prompts and document sets for each workload.
CLIENT	Ollama/OpenAI-compatible adapter wraps model calls and response logs.
ENERGY	NVML samples GPU power/utilization/VRAM; CPU uses RAPL or labeled TDP estimate.
VISUALS	pandas + matplotlib turn traces into poster-ready figures.

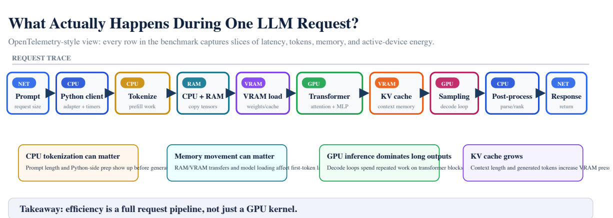
> **Energy attribution matters:** active-device energy is cleaner for model comparison; total wall energy is better for full-machine cost.

WHAT GETS TIMED

Slice	Meaning	Why Python Devs Care
Network	client call overhead	endpoint placement, batching
Tokenization	prompt processing / prefill	long docs are not free
Inference	model generation	model size dominates
Post-processing	parse, serialize, rank	pipeline glue is measurable

WHAT ACTUALLY HAPPENS DURING ONE LLM REQUEST?

Takeaway: efficiency is a full request pipeline, not just a GPU kernel — FULL PIPELINE



A slow or expensive call can come from tokenization/prefill, memory movement, KV cache growth, GPU inference, sampling, networking, or response parsing.

WHAT THE NUMBERS MEAN

Toy task score
Small reproducible task signal. Example: 0.67 means 2 of 3 deterministic checks passed.

Energy
Active-device joules: GPU NVML draw minus idle baseline, or labeled CPU estimate.

Tokens/J
Generated or processed tokens divided by active-device energy.

Interpretation
Use trends for deployment decisions; this is not a universal model leaderboard.

LEARNING 1: BIGGER MODELS HAVE D

SAME ARCHITECTURE · DIFFERENT SIZES

Scaling parameters often increases infrastruct

Same-family comparisons isolate parameter count: Gemma 2B/

Measured benchmark: toy score vs infrastructure c

> **Score note:** the quality panel is a tiny code-generat

LEARNING 2: QUANTIZATION CHANGE

SAME MODEL · DIFFERENT PRECISION

4-bit and 8-bit models often retain useful qual

Measured benchmark: fp16 vs q8 vs q4 only within

> Quantization changes representation efficiency, not

Artifacts: benchmark framework, configs, notebooks, CSV metrics, NVML traces, generated PNG/SVG assets, and printable poster.
Open repository: github.com/shivaylamba/pycon-2026-poster | **Inspired by:** Alizadeh et al., *Language Models in Software Development Tasks: An Experimental Analysis of Energy and Accuracy*, arXiv:2412.00329.